DEPARTMENT OF SOFTWARE ENGINEERING

Assignment # 2 (Infinite Series CLO-2)

COURSE CODE:MT-224

STUDENT NAME	ROLL NUMBER
RAAZIA	SE-24009

Question 1: Total Distance of a Bouncing Ball

Problem Statement: A ball is dropped from a height of 10 m. Each time it strikes the ground, it bounces vertically to a height that is 3/4 of the preceding height. Find the total distance the ball travels if it bounces infinitely often.

Reference Formula: For a bouncing ball, the total distance D is given by:

$$D = H + 2\sum_{n=1}^{\infty} ar^n = H\left(\frac{1+r}{1-r}\right)$$

Solution:

1. **Identify Given Values:** Initial height $(H) = 10$ m
   Common ratio $(r) = 3/4 = 0.75$

2. **Setup the Series:** The ball falls 10 m first. Every bounce thereafter involves an upward and downward trip.

$$D = 10 + 2\left[10\left(\frac{3}{4}\right) + 10\left(\frac{3}{4}\right)^2 + \dots\right]$$

3. **Apply Infinite Sum:** The term in brackets is an infinite geometric series where $a = 7.5$ and $r = 0.75$.

$$S = \frac{a}{1-r} = \frac{7.5}{1-0.75} = \frac{7.5}{0.25} = 30$$

4. **Calculate Final Distance:**

$$D = 10 + 2(30)$$

$$D = 10 + 60 = 70 \text{ m}$$

Final Result: The total distance traveled is **70 m**.

Question 2: Repeated Squaring Sequence

Problem Statement: A student enters 0.5 in a calculator and repeatedly squares the number. Find the general term a_n, the limit, and the values of a_0 for convergence.

Solution:

(i) General Term Formula: The recurrence is $a_{n+1} = (a_n)^2$.

- $a_0 = 0.5$
- $a_1 = (a_0)^2$
- $a_2 = (a_1)^2 = ((a_0)^2)^2 = (a_0)^4$
- $a_3 = (a_2)^2 = ((a_0)^4)^2 = (a_0)^8$

By observation, the exponent follows the sequence 2^n. Therefore:

$$a_n = (a_0)^{2^n} = (0.5)^{2^n}$$

(ii) Limit Analysis: We evaluate the behavior as $n \to \infty$:

$$\lim_{n\to\infty} (0.5)^{2^n}$$

Since $2^n \to \infty$ and the base $|0.5| < 1$, the terms exponentially approach zero.

$$\text{Limit} = 0$$

(iii) Convergence Conditions:

- **Case $|a_0| < 1$:** Sequence converges to 0.
- **Case $|a_0| = 1$:** Sequence converges to 1 (since $(\pm 1)^2 = 1$).
- **Case $|a_0| > 1$:** Sequence diverges to ∞.

Conclusion: The procedure produces a convergent sequence for $|a_0| \leq 1$.

Question 3: Image Compression Information Value

Problem Statement: Calculate the maximum possible compressed image value if the first term is 100 and each subsequent term is 75% of the previous one.

Reference Formula: Sum of an infinite geometric series: $S = \frac{a}{1-r}$

Step-by-Step Solution:

1. **Identify Values:** First term $(a) = 100$
   Common ratio $(r) = 75\% = 0.75$
2. **Check Convergence:** Since $|r| = 0.75 < 1$, the series converges to a finite limit.
3. **Substitute and Solve:**

$$S = \frac{100}{1 - 0.75}$$
$$S = \frac{100}{0.25}$$
$$S = 400$$

Final Result: The maximum possible compressed image information value is **400 units**.

Question 4: Data Packet Size Reduction

Problem Statement: A software bug causes data packets to reduce in size by 10% each time, starting with an initial size of 1 MB. Find the total data transmitted.

Step-by-Step Solution:

1. **Identify Given Values:** Initial size (a) = 1 MB
   Reduction = 10%, therefore each packet is 90% of the previous ($r = 0.9$).

2. **Formulate the Series:** The total data is the sum: $1 + 0.9 + 0.9^2 + 0.9^3 + \dots$

3. **Calculate Sum:** Using $S = \frac{a}{1-r}$:

$$S = \frac{1}{1 - 0.9}$$

$$S = \frac{1}{0.1}$$

$$S = 10 \text{ MB}$$

Final Result: The total data transmitted converges to **10 MB**. Since the result is finite, the system remains stable.

Question 5: Poles and Zeroes Analysis

Part A: Impedance Function

$$Z(s) = \frac{s+2}{(s+1)(s+3)}$$

- **Zeroes:** Set numerator to zero.

$$s + 2 = 0 \implies \mathbf{s = -2}$$

- **Poles:** Set denominator to zero.

$$(s+1)(s+3) = 0 \implies \mathbf{s = -1, -3}$$

Part B: Digital Filter

$$H(z) = \frac{z-1}{z^2 - 0.5z - 0.6}$$

- **Zeroes:**

$$z - 1 = 0 \implies \mathbf{z = 1}$$

- **Poles:** Solve $z^2 - 0.5z - 0.6 = 0$ using the quadratic formula $z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$:

$$z = \frac{0.5 \pm \sqrt{(-0.5)^2 - 4(1)(-0.6)}}{2}$$

$$z = \frac{0.5 \pm \sqrt{0.25 + 2.4}}{2} = \frac{0.5 \pm 1.628}{2}$$

$$\mathbf{z \approx 1.064, -0.564}$$

Question 6: System Stability Analysis

Problem Statement: Identify poles and zeroes of $H(z) = \frac{z^2-5z+6}{z^2-3z+2}$ and determine stability.

1. Find Zeroes: Set $z^2 - 5z + 6 = 0$. Factorizing: $(z-2)(z-3) = 0$. **Zeroes at** $z = 2, 3$.

2. Find Poles: Set $z^2 - 3z + 2 = 0$. Factorizing: $(z-1)(z-2) = 0$. **Poles at** $z = 1, 2$.

3. Stability Check: For a discrete-time system, stability requires all poles to be inside the unit circle ($|z| < 1$).

- Pole at $z = 1$: $|1| = 1$ (Marginally stable/on the boundary).
- Pole at $z = 2$: $|2| = 2$ (Outside the unit circle).

Conclusion: Because the pole at $z = 2$ lies outside the unit circle, the system is **unstable**.